PROJECT REPORT · NOVEMBER 2026

CLONEGUARD

Claim-routed voiceprint verification and Hinglish scam-pattern analysis for impersonation calls.

Built on Gnani AI — Prisma · Voice-clone embeddings · Timbre

CREATORS

KrushKadam — github.com/KrushKadam
Legendry-Sannin — github.com/Legendry-Sannin

BUILT FOR

Gnani AI — Great Indian AI Internship Challenge
#GnaniAI #GreatIndianAIInternshipChallenge

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CLONEGUARD — PROJECT REPORT

A two-layer check for impersonation phone calls: voiceprint identity matching plus Hinglish scam-pattern analysis

Authors / creators	**KrushKadam** — https://github.com/KrushKadam · **Legendry-Sannin** — https://github.com/Legendry-Sannin
Team	Two-member team, joint creators of CloneGuard
Built for	Gnani AI — Great Indian AI Internship Challenge (deadline 10 November 2026) · `#GnaniAI` `#GreatIndianAIInternshipChallenge`
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ABSTRACT

Phone scams in India often follow one script: a caller claims to be a parent, a boss or an official, invents an emergency, and demands money urgently and in secret. The script is usually spoken in Hinglish (Hindi written in English letters and mixed with English), and voice cloning makes the claimed identity cheaper to fake. CloneGuard checks a call recording two ways. **Layer 1** compares the caller's voice fingerprint (a 768-number speaker embedding from Gnani's voice-clone service) with the enrolled voiceprint of the person the caller *claims* to be, which it picks from spoken claims such as "Main Papa bol raha hoon". The comparison uses cosine similarity against a threshold of 0.55. **Layer 2** transcribes the call with Gnani Prisma and scans it for 32 Hinglish red-flag patterns (204 variants in Latin and Devanagari script) across six categories. The two risks combine with equal weights into a 0–100 score with a Low, Caution or High band and a plain-English verdict, which Gnani Timbre can read aloud. Each check is logged locally and can be exported as a PDF evidence report. A built-in Hinglish scam-call generator and a Self-Test Harness supply labeled test calls and a calibration scorecard. On ten builder-written scenarios the harness caught 7/7 scams at High and passed 3/3 genuine calls at Low, with no Caution results and no errors. The result was the same at every threshold from 0.45 to 0.65 and was reproduced on a fresh Linux machine. The interface runs locally; the analysis runs on Gnani's cloud APIs.

Keywords: speaker verification, voice impersonation, vishing, social engineering, Hinglish, code-mixed speech, scam detection, explainable risk scoring, Gnani AI.

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1. INTRODUCTION AND MOTIVATION

A call comes in. The voice says *"Beta, main Papa bol raha hoon. Mera accident ho gaya hai..."* ("Son, it's Papa. I've had an accident..."). Money is needed *turant* (right now), and *kisi ko abhi mat batana* (don't tell anyone yet). The person receiving the call has two questions and very little time to answer them:

1. **Is this really the person they claim to be?**
2. **Does this conversation follow a known scam pattern?**

Voice cloning and ordinary human impersonation both make the first question harder to answer by ear. English-language scam filters miss the second, because the important words are Hinglish. CloneGuard answers both questions at once and shows its evidence so that a person, not the tool, makes the final decision.

2. PROBLEM STATEMENT AND OBJECTIVES

Problem. Given a recording of a call and a set of trusted people whose voices have been enrolled, estimate the risk that the call is an impersonation scam. Show the evidence behind that estimate in plain language.

Objectives.

#	OBJECTIVE	HOW IT IS MET
01	Compare the caller with the person they *claim* to be, not a generic voice	Claim detection selects the enrolled voiceprint (Section 6.4)
02	Detect Hinglish scam language in either script	32 dual-script patterns across six categories (Section 6.3)
03	Produce one explainable score	Equal-weight composite with visible arithmetic and three bands (Section 6.5)
04	Be testable without real victims or a microphone	Built-in generator and Self-Test Harness (Sections 7.3–7.4)
05	Leave a shareable, dated record	Incident log and per-check PDF evidence report (Section 7.5)
06	Be honest about limits	Four limitations shown in the UI, the PDFs and this report (Section 12)

3. BACKGROUND AND RESEARCH REFERENCES

CloneGuard draws on four research threads. Full citations are in Section 18. Each was verified by fetching the source on 4 November 2026 (see `docs/01_PRIOR_ART.md`, Section 7).

3.1 Speaker embeddings. Modern speaker verification represents a voice as a fixed-length vector and compares vectors with cosine similarity. ECAPA-TDNN [R1] is a well-known architecture of this kind. CloneGuard uses Gnani's 768-dimension voice-clone embeddings. We do not claim that Gnani uses ECAPA-TDNN; the reference explains the general technique.

3.2 Spoofing and its separation from verification. The ASVspoof challenge series [R3, R9] and the tandem detection cost function (t-DCF) [R2] treat *speaker verification* ("is this the enrolled person?") and *spoofing countermeasures* ("is this audio synthetic or replayed?") as separate tasks that must be evaluated together. CloneGuard's Layer 1 is a verification signal only. For that reason it reports an **identity mismatch**, never "synthetic audio".

3.3 Indian-language speech. IndicSUPERB [R4] provides an Indian-language speech benchmark. IndicSynth [R5] provides about 4,000 hours of synthetic speech in 12 Indian languages and shows that detectors trained only on English generalise poorly. Gaikawad and Ghosh [R6] report a lightweight CNN-Transformer for audio deepfake detection in six Indian languages. These works motivate Indian-language evaluation and are the basis for the future work in Section 14.

3.4 Scam-language analysis. VISHGUARD [R7] is a dataset of 3,000 simulated vishing (voice phishing) calls annotated for persuasion strategy and emotional tone. It is the closest research analogue to Layer 2, but it contains no Indian languages. Meena et al. [R8] detect phishing in Hinglish text using urgency, emotional and call-to-action cues, a direct precedent for treating Hinglish text patterns as a fraud signal. Surveys of voice cloning [R10] and of generated-voice detection [R11] confirm that the surrounding literature is established.

4. EXISTING SOLUTIONS

SOLUTION	TYPE	WHAT IT DOES	HOW CLONEGUARD DIFFERS
Pindrop (Pulse, Passport, VeriCall, Protect) [S1]	Enterprise	Identity trust for contact centres: synthetic-speech detection, caller authentication, phone-number (ANI) validation, live fraud scoring	Desktop tool for households and small teams; no sales contract or integration; shows its arithmetic
Resemble AI (Detect, Identity, Intelligence) [S2]	Enterprise	Multimodal deepfake detection across 54 languages; an Identity API (documented as beta) that enrolls and matches voices	Adds a Hinglish scam-language layer next to identity, picks the comparison target from the spoken claim, and includes self-calibration
Reality Defender [S3]	Enterprise	Deepfake detection platform (reported by a second sweep; site unreachable during ours)	—
Tring [S4]	Consumer, India	AI call assistant in 12+ Indian languages including Hinglish: spam blocking, live transcripts, answering in the user's own cloned voice	Verifies a *claimed* identity against an enrolled trusted person instead of creating a voice
VoxDetect [S5]	Open source	Prototype that combines a Wav2Vec2 deepfake classifier, prosody, voiceprint and call context into a 0–100 score	Hinglish dual-script pattern set, a built-in scam-call generator, a self-test scorecard, and per-check PDF evidence
SIH26104 cohort [S6]	Hackathon	Smart India Hackathon 2026 problem: real-time detection and prevention of voice-cloning impersonation; 28+ related public repositories, mostly early-stage	A finished, measured, documented end-to-end build with calibration evidence

A partial search cannot prove that any feature is absent from the market. The differences above are at the feature level and limited to the public sources reviewed.

5. INNOVATION — WHAT CLONEGUARD ADDS

The individual parts (speaker embeddings, keyword scans, risk scores) are known. What this project adds is how they are **combined, targeted and checked**:

1. **Claim-routed identity verification.** The transcript chooses which enrolled voiceprint to compare against. "Main Papa bol raha hoon" compares the caller with *Papa*, and "CEO bol raha hoon" compares with the *CEO*. A claim that names someone not enrolled is reported on screen instead of being silently compared with the wrong person.
2. **Two independent layers that stay visible.** Voice identity and conversation language are scored separately, and both remain on screen with the threshold, the matched phrases and their snippets. The composite is a simple, explainable formula rather than an opaque classifier.
3. **Hinglish scam patterns written for code-mixed speech, in both scripts.** There are 32 patterns and 204 variants across urgency, OTP/bank, authority, threats, emotional pressure and call-to-action.

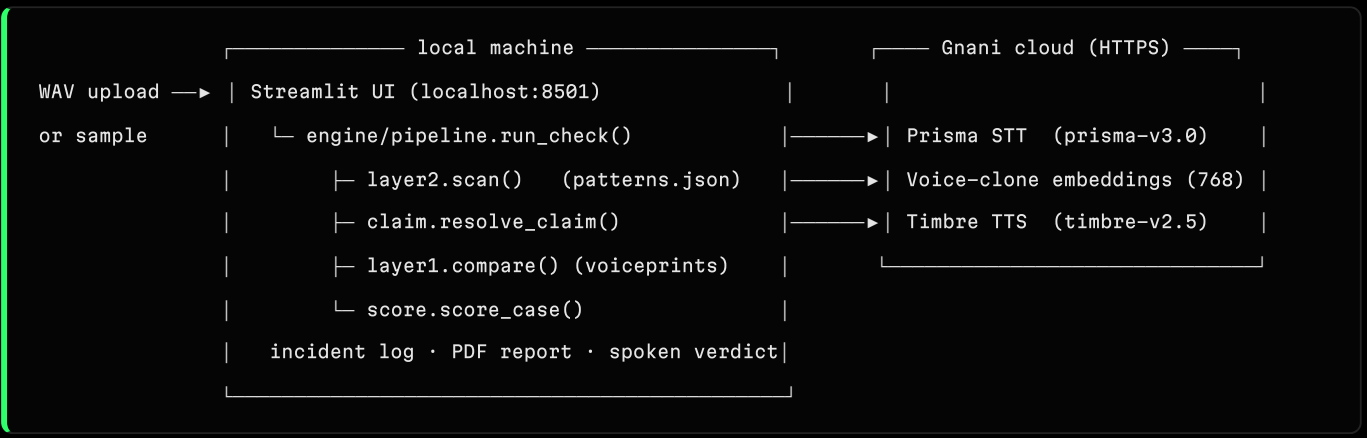
Each is stored in Latin and Devanagari script, so the same rules work whichever script the speech-to-text returns. A 51-variant subset returned zero hits on the genuine transcripts.

4. **Self-attacking calibration.** A built-in generator produces scam and genuine calls with known labels (Timbre voices plus a phone-line effect). The Self-Test Harness runs them through the full pipeline and publishes a scorecard, including the weakest scenario and a threshold sweep. The scorecard prints its own caveat: calibration against builder-written scripts is not independent accuracy.
5. **Honest output by design.** The tool never says "AI-cloned" or "safe". Layer 1 reports an identity match or mismatch, Low risk is called "a weak signal, not a clearance", and the four limitations appear in the UI, every PDF and this report.
6. **Evidence you can hand over.** A dated local incident log and a per-check PDF evidence report (native Devanagari) make the result shareable without re-running any analysis.
7. **Composition of three Gnani services into one local verdict.** Prisma (speech-to-text), the voice-clone embedding endpoint (speaker fingerprints) and Timbre (spoken verdict and test-call generation) work together, with caching so that re-checking unchanged audio costs no API credits.

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6. SYSTEM DESIGN

6.1 Architecture



6.2 Layer 1 — voiceprint match

- The caller's audio is turned into a 768-number embedding by Gnani's voice-clone embedding endpoint. Because the vector arrives as a string, `voiceprint.py` parses JSON, CSV, tensor-repr and base64 float32/bfloat16 formats defensively.
- Cosine similarity against the target voiceprint gives **match** at 0.55 or above and **mismatch** below it.
- `layer1_risk = clamp((0.55 − cosine) / 0.55, 0, 1)`

6.3 Layer 2 — Hinglish scam-pattern scan

- The transcript is normalised: NFKC, lowercase, punctuation removed, zero-width characters dropped, whitespace collapsed. Devanagari combining marks are preserved.
- Substring matching runs against the six categories: urgency (5 patterns), OTP/bank (5), authority (5), threats (6), emotional (5), call-to-action (6). Each hit records its category, phrase, snippet, severity

and weight.

- `layer2_risk = min(1, Σ hit weights / 0.8)`

6.4 Claimed-identity routing

Exact lookups for Papa, Mummy and CEO, in Latin and Devanagari variants, run over the same normalised text. There are three outcomes:

- **Claim, enrolled:** compare with that voiceprint.
- **Claim, not enrolled:** the screen says "claim names someone not enrolled", and the score comes from Layer 2 alone, rescaled to 0–100.
- **No claim:** compare with the default voiceprint, and the screen says so.

6.5 Composite score and bands

`composite = round(100 × (0.5 × layer1_risk + 0.5 × layer2_risk))`

BAND	RANGE	VERDICT SENTENCE
Low	0–29	"The caller's voice is consistent with the enrolled person and no scam patterns matched. This is a weak signal, not a clearance."
Caution	30–59	"One layer raised a concern. Ask a verification question the real person would answer."
High	60–100	"The caller does not match the claimed person's voice, or the conversation uses known scam patterns, or both. Treat this call as unsafe."

7. IMPLEMENTATION

7.1 Technology stack

Python · Streamlit 1.66.1 (local UI) · Gnani Prisma, voice-clone embeddings and Timbre · fpdf2 2.9 with uharfbuzz and fontTools (Devanagari PDF shaping) · pypdf (PDF verification) · requests · JSON file storage.

7.2 Modules

PATH	RESPONSIBILITY
`app.py`	Streamlit UI: Check a call, Spoken verdict, Enroll a voice, Self-test, Incident log; matrix-green design system
`config.py`	Thresholds, weights, band cut-offs, model names, paths
`gnani.py`	Prisma and Timbre client
`voiceprint.py`	Embedding retrieval, defensive parsing, cosine
`engine/pipeline.py`	`run_check()` : validate → transcribe → scan → claim → compare → score; caching; incident write

PATH	RESPONSIBILITY
`engine/layer1.py`, `layer2.py`, `claim.py`, `score.py`	The two layers, claim routing, scoring
`generator/`	Hinglish scam and genuine scripts, Timbre voicing, phone-line effect
`selftest/`	Harness, threshold sweep, scorecard PDF
`reports/incident_pdf.py`	Per-check A4 evidence report with native Devanagari
`security/`	Threat model, attack surface, secrets policy, checklist, audit

7.3 Scam-call generator

Ten scenarios were generated: seven scams voiced by Timbre "Deepak" (family emergency, bank fraud, executive, government/police, tech support, prize refund, and a scam with no urgency words) and three genuine calls voiced by "Nalini" (friend dinner, relative check-in, and a genuine call that mentions money). The build machine had no microphone, so every test file comes from the generator.

7.4 Self-Test Harness

The harness runs every scenario through `run_check()`. A scam counts as caught only at High, and a genuine call passes only at Low; Caution counts as a miss in both directions. It then re-scores the stored results at thresholds 0.45, 0.50, 0.55, 0.60 and 0.65, names the weakest scenario, and exports a scorecard PDF. Per-scenario API failures are recorded as errors and never crash the run.

7.5 Evidence and logging

Every check is appended to `data/incidents.json` before any PDF is made, so a failed report cannot lose the check. The PDF contains both layer scores, the matched phrases, a transcript excerpt, the verdict and the four limitations. No audio is stored in the log.

8. METHODOLOGY AND TEST DATA

- **Ground truth:** builder-written scripts with labels known at generation time (`data/scenarios.json`).
- **Enrollment:** a voice sample (`samples/A1_nalini_X.wav`) is enrolled with the role "Papa".
- **Hard cases:** a genuine call that mentions money (tests false positives) and a scam without urgency words (tests reliance on a single category).
- **Voice separation:** `voiceprint_test.py` measures same-speaker and different-speaker cosine values. Embeddings are deterministic, so the same audio always gives the same numbers.
- **Metrics:** caught/total for scams, passed/total for genuine calls, Caution count, error count, per-scenario distance from the expected band, and threshold sensitivity.

9. RESULTS

9.1 Voice separation

COMPARISON	COSINE
Genuine call vs Papa voiceprint (end-to-end run)	**0.8263**
Same speaker, two clips	**0.7306**
Different speakers	**0.2964**
Scam call vs Papa voiceprint (end-to-end run)	**0.2752**
Separation gap (same – different)	**0.4342**
Threshold used	**0.55**

9.2 Calibration scorecard

#	SCENARIO	EXPECTED	COMPOSITE	BAND	VERDICT
1	family_emergency	scam	75	high	caught
2	bank_fraud	scam	74	high	caught
3	executive	scam	75	high	caught
4	government_police	scam	76	high	caught
5	tech_support	scam	74	high	caught
6	prize_refund	scam	76	high	caught
7	friend_dinner	genuine	0	low	passed
8	college_exam_fee *(hard case)*	genuine	0	low	passed
9	relative_checkin	genuine	0	low	passed
10	bank_refund_no_urgency *(hard case)*	scam	71	high	caught

Headline: scam caught **7/7**, genuine passed **3/3**, Caution **0**, errors **0**. The threshold sweep from 0.45 to 0.65 gave 7/7 and 3/3 at every setting. The weakest scenario was family_emergency (composite 75, distance 0). The score weights stayed at 0.5/0.5 because calibration gave no evidence for moving them. Sources: run `run_20261107_183026_08f8` (build machine), reproduced as `run_20261110_082347_ef50` (fresh Linux machine).

9.3 Worked example: the family-emergency scam

The caller says "Beta, main Papa bol raha hoon...", so the claim is detected and Papa's voiceprint is selected. The cosine is 0.2752, below 0.55, which gives **identity mismatch** and a Layer 1 risk of 0.50. Layer 2 finds four 0.20-weight hits: तुरंत (urgency), एक्सीडेंट हो गया है (emotional), किसी को अभी मत बताना (emotional) and पैसे भेज दो (call-to-action), so the Layer 2 risk is 1.00. The composite is 100 × (0.5 × 0.50 + 0.5 × 1.00) = **75** → **High risk**.

10. FRESH-MACHINE END-TO-END VALIDATION

On 10 November 2026 the repository was cloned onto a clean Kali Linux machine (Python 3.13.15, commit `c3c50e8`) with no voiceprints and no caches, and tested through the real UI in a headless browser:

PHASE	RESULT
Preflight: key present, endpoint reachable (422 probe, no credit)	PASS
Static: 21 Python files compile; wording scan; palette scan (0 illegal colours)	PASS
UI smoke: 5 pages, 0 exceptions, HTTP 200	PASS
Enroll Nalini as Papa	PASS
Scam check → 75, High; genuine check → 0, Low	PASS
Claim routing: CEO not enrolled → note shown; CEO enrolled (CCO third-party voice) → routed	PASS
Evidence PDF and scorecard PDF readable; spoken verdict returned	PASS
Self-test: 7/7, 3/3, 0, 0, sweep flat	PASS

The session used 27 Gnani API calls and was recorded as a demo video with the app's own audio.

Defects found and documented, not yet fixed:

1. The scorecard PDF says "Nirmala UI (C:\Windows\Fonts\Nirmala.ttc) is installed on this machine" even on Linux.
2. An enrolled voiceprint's "Source" shows the temporary upload path instead of the original file name.

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11. THREATS TO VALIDITY

- **Builder-written ground truth.** The scenarios were written by the same team that wrote the patterns, so the scorecard measures calibration, not independent accuracy. The scorecard says this itself.
- **Small sample.** Ten scenarios with two Timbre voices. The flat threshold sweep shows the result is robust on this set, but not that it generalises.
- **Synthetic speech only.** All calls are text-to-speech with a simulated phone-line effect. There are no real human callers, real telephone channels, background noise or compression.
- **Same engine family.** Enrollment and test calls both come from Timbre voices. Real-world voices and microphones will vary more.
- **Exact-substring matching.** Paraphrased scam language that avoids the listed phrases will not match (Limitation 4).

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12. LIMITATIONS

1. **A perfect clone of the enrolled voice passes Layer 1.** Matching an identity is not proof that the audio is genuine.

2. **A mismatch is not proof of synthetic audio.** Layer 1 flags an identity mismatch. That catches clones and human impersonators alike.
3. **The analysis runs on Gnani's cloud.** Audio leaves the machine, and the app needs an internet connection. It is a local interface with cloud inference.
4. **A partial search cannot prove absence.** "No patterns matched" means the phrase list did not match. It does not mean the call is safe.

13. ETHICS, PRIVACY AND SECURITY

- **Decision support, not judgment.** The verdict is a risk signal with evidence. A human makes the final decision, and the UI says so.
- **No real victims or voices in testing.** Test calls are generated, and the fresh-machine test used a CC0 open-source TTS sample from Wikimedia Commons as its unknown third-party voice.
- **Consent.** Enroll only voices of people who have agreed, and treat voiceprints as personal data.
- **Data stays local.** Voiceprints, incidents and caches live in `data/` (gitignored). Only inference requests go to Gnani over HTTPS.
- **Secrets.** `GNANI_API_KEY` is read from the environment and never written to disk or logs.
- **Security audit (9 November 2026).** Generated data was untracked from git, uploads were capped at 25 MB with a WAV header check, sample paths were guarded, and dependencies were pinned. There is no eval, pickle or exec, no secrets in history, and no 0.0.0.0 binding by default. Local-only breachability was rated LOW. See `security/`.

14. FUTURE WORK AND RESEARCH DIRECTIONS

Engineering. A desktop background monitor for user-supplied audio; fixes for the two defects in Section 10; fuzzy and paraphrase-tolerant Layer 2 matching; more claim roles.

Research questions for a follow-up paper.

RQ	QUESTION	PROPOSED EVALUATION
RQ1	How well does claim-routed verification separate impersonators from enrolled speakers on real Indian telephone speech?	Equal error rate (EER) on recorded and consented call audio; IndicSUPERB-style speaker sets [R4]
RQ2	Does adding a spoofing countermeasure next to Layer 1 reduce the "perfect clone passes" risk (Limitation 1)?	t-DCF-style tandem evaluation [R2] on ASVspoof [R3, R9] and IndicSynth [R5]
RQ3	How well do dual-script Hinglish patterns generalise compared with a learned classifier?	Held-out, independently written scripts; comparison against an ML baseline in the style of [R8]
RQ4	What weights and threshold are best when tuned on independent data?	Cross-validated tuning of W_L1, W_L2, the divisor and the threshold, replacing the starting values
RQ5	Does an explainable two-layer verdict help people decide better than a single score?	User study with consent and ethics approval

15. AUTHORSHIP, INTELLECTUAL PROPERTY AND PUBLICATION RIGHTS

15.1 Authorship. CloneGuard — its design, source code, pattern set, test scenarios, documentation, evaluation and media — was created jointly by:

- **KrushKadam** — https://github.com/KrushKadam
- **Legendry-Sannin** — https://github.com/Legendry-Sannin

as a team for the Gnani AI Great Indian AI Internship Challenge.

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15.4 Dated evidence of authorship.

- Git history, 5–10 November 2026: 24 dated commits from `4ac0838` (pre-build documentation, 5 Nov) to `c3c50e8` (10 Nov). The commit author is recorded as Krushna Kadam (KrushKadam).
- Dated design, requirements and innovation documents in `docs/`, written before and during the build.
- Recommended: timestamp a zip of the repository with OpenTimestamps (free; see `docs/00_PROJECT_INDEX.md`, Section 6) and keep the challenge submission confirmation.

15.5 Things to check before publishing.

- **Challenge terms.** Read the Gnani AI challenge's official rules for any intellectual-property, licensing or publicity clauses. This report does not know those terms, and they take precedence over anything stated here.
- **Patents.** Public disclosure (GitHub, LinkedIn, video) before a patent filing can affect patentability in India and elsewhere. Research papers are not affected. If a patent is ever wanted, get advice from a registered patent agent before any further disclosure.
- **Third-party terms.** Results depend on Gnani's APIs, which are governed by Gnani's terms of use. Dependencies keep their own licences: Streamlit (Apache-2.0), requests (Apache-2.0), fpdf2 (LGPL-3.0-only), pypdf (BSD-3-Clause), fontTools (MIT), uharfbuzz (Apache-2.0). The fonts used in the media (Chakra Petch, Martian Mono, Noto Sans Devanagari) are under the SIL Open Font License.
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- *This section is a practical summary, not legal advice.*

15.6 How to cite

```
@misc{cloneguard2026,
  author       = {KrushKadam and Legendry-Sannin},
  title        = {CloneGuard: Claim-Routed Voiceprint Verification and Hinglish Scam-Pattern Analysis for Impe
  year         = {2026},
  month        = nov,
  howpublished = {\url{https://github.com/KrushKadam/CloneGuard}},
  note         = {Gnani AI Great Indian AI Internship Challenge}
}
```

16. REPRODUCIBILITY GUIDE

```
git clone https://github.com/KrushKadam/CloneGuard.git && cd CloneGuard
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
export GNANI_API_KEY="<your key>"        # never commit it
python3 READY_CHECK.py                    # 422 probe: key + endpoint, no credits
streamlit run app.py                      # http://localhost:8501
# Enroll samples/A1_nalini_X.wav as Papa, then:
python3 selftest/harness.py               # calibration scorecard
```

On a fresh machine the battery makes about 2 API calls per uncached scenario. Re-runs on cached audio make none.

17. PROJECT TIMELINE

DATE (2026)	MILESTONE	COMMIT
5 Nov	Pre-build documentation (index, prior art, positioning, requirements, architecture, data model, innovation report)	`4ac0838`
5 Nov	Build step 1: Hinglish scam-call generator	`8e1d710`
5 Nov	Build step 2: detection engine (both layers, claim, score, pipeline)	`abd10ee`
5 Nov	Build step 3: Self-Test Harness and scorecard PDF	`1cb4cfb`
6 Nov	Build step 4: Streamlit UI	`34207bd`
7 Nov	Build step 5: per-check PDF evidence report	`206e08c`
7 Nov	Calibration: 32 dual-script patterns, 7/7 and 3/3	`8c2b76e`
8 Nov	Documentation pass, README, contest pack, final certification	`8b576b9`, `12cd207`, `290f946`

DATE (2026)	MILESTONE	COMMIT
9 Nov	Plain-English UI pass; security hardening	`8d0deb6`, `1cdeb5a`
10 Nov	Security docs; UI design system; artifacts bundle; E2E prompt	`2bcd980` … `c3c50e8`
10 Nov	Fresh-machine end-to-end test, demo video, project overview PDF, this report	(working tree)

18. REFERENCES

Research

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- **[R5]** Sharma, B., Ekbote, N., & Gupta, A. (2025). IndicSynth: Evaluation of Synthetic Speech Across Indian Linguistic Accents and Code-Switching. *ACL 2025 (Long Papers)*, 22037–22060. https://doi.org/10.18653/v1/2025.acl-long.1070
- **[R6]** Gaikawad, S., & Ghosh, S. (2025). A Robust and Lightweight CNN-Transformer Model for Audio Deepfake Detection in Indian Languages. *ISPCC 2025*, 382–387. https://doi.org/10.1109/ISPCC66872.2025.11039572
- **[R7]** Hmimou, Y., Tabaa, M., Khiat, A., & Hidila, Z. (2026). A Multilingual Behavioral Speech Dataset for Vishing and Social Engineering Detection (VISHGUARD). *Scientific Data*, 13, 1456. https://doi.org/10.1038/s41597-026-07724-z
- **[R8]** Meena, D., Dayanvita, Singh, M., Shweta, & Singh, J. (2024). Hinglish Text Phishing Detection Model Using Machine Learning Algorithms. *IGDTUW Research Showcase, Compendium of Abstracts*, abstract 43. https://www.igdtuw.ac.in/IGDTUW/uploads/919646825.pdf *(A research-showcase abstract, not a separately verified peer-reviewed paper.* `docs/01_PRIOR_ART.md` *also lists a related Taylor & Francis chapter, DOI 10.1201/9781003731689-104; confirm which version to cite before submitting a paper.)*
- **[R9]** Liu, X., et al. (2023). ASVspoof 2021: Towards Spoofed and Deepfake Speech Detection in the Wild. *IEEE/ACM TASLP*. https://doi.org/10.1109/TASLP.2023.3285283 · https://arxiv.org/abs/2210.02437
- **[R10]** Voice Cloning: Comprehensive Survey (2025). https://arxiv.org/abs/2505.00579

- **[R11]** A survey of AI-generated voices and their detection (2025). *Emerald*. https://doi.org/10.1108/ATSIP-09-2025-0093

Existing solutions and sources

- **[S1]** Pindrop — https://www.pindrop.com/ · https://www.pindrop.com/pulse-for-contact-centers
- **[S2]** Resemble AI — https://www.resemble.ai/ · https://www.resemble.ai/products/detect
- **[S3]** Reality Defender (via Resemble's comparison page) — https://www.resemble.ai/resemble-ai-vs-reality-defender *(second-sweep report, not independently fetched)*
- **[S4]** Tring — https://heytring.com/
- **[S5]** VoxDetect — https://github.com/io-PEAK/VoxDetect
- **[S6]** SIH26104 problem statement — https://github.com/vedantchalke36/sih-2026-problem-statements/blob/main/ps_2026/SIH26104.md · https://www.sihbuddy.in/ps/SIH26104
- **[S7]** ASVspoof challenge series — https://www.asvspoof.org/

Platform

- **[P1]** Gnani AI APIs (Prisma speech-to-text `prisma-v3.0`, voice-clone embeddings, Timbre text-to-speech `timbre-v2.5`) via `https://api.vachana.ai`; usage page at app.gnani.ai.

Unless marked otherwise, all sources were verified by direct fetch on 4 November 2026 (`docs/01_PRIOR_ART.md` §7, `docs/06_INNOVATION_REPORT.md` §10).

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19. APPENDICES

A. Configuration constants (`config.py`): `THRESHOLD = 0.55`, `W_L1 = W_L2 = 0.5`, `LAYER2_DIVISOR = 0.8`, `BAND_CUTOFFS = (30, 60)`, `STT_MODEL = "prisma-v3.0"`, `LANGUAGE_CODE = "indic-auto"`.

B. Pattern categories (`data/patterns.json`): urgency 5 · OTP/bank 5 · authority 5 · threats 6 · emotional 5 · call-to-action 6 = 32 patterns, 204 Latin and Devanagari variants.

C. Project documents: `docs/00` – `08` (index, prior art, positioning, requirements, architecture, data model, innovation report, contest pack, E2E prompt), `security/`, `artifacts/`.

D. Deliverables produced on 10 November 2026: a 60-second narrated LinkedIn demo video with real call audio (`brag-output/brag.mp4`), a 25-page themed project overview (`brag-output/CloneGuard-Project-Overview.pdf`), the full end-to-end test recording (`DEMO_ASSETS/video/`), and this report.

E. Glossary. *Embedding / voiceprint:* a list of numbers describing how a voice sounds. *Cosine similarity:* a 0-to-1 score where 1 means identical direction (voices). *Hinglish:* Hindi written in Latin letters, mixed with English. *Vishing:* voice phishing. *EER:* equal error rate. *t-DCF:* tandem detection cost function. *STT / TTS:* speech-to-text / text-to-speech.

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